Part 1: A Statistical Modeling Approach to Predicting Residential Property Values in Ames, Iowa

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Introduction and Description of the Data:

To begin, the goal of this project was to develop a regression model that predicts residential sale prices in Ames, Iowa using key structural and lot characteristics. The data was drawn from a subset of the Ames Housing dataset, which includes information on square footage, lot size, construction quality, basement features, garage characteristics, and exterior amenities. Knowing that housing prices typically change in percentages rather than by fixed dollar amounts, we modeled the natural logarithm of SalePrice instead of the raw price. This improves the model's fit and allows the predicted effects to be interpreted as approximate percentage changes, which is more meaningful in real estate decision-making. However, before estimating the model, highly influential observations were removed using Cook's Distance, resulting in a final sample of 927 homes. Several variables were also constructed to better capture property features. For example, Age was calculated as the difference between the year sold and the year built. Similarly, TotalPorchSF combines all porch-related square footage, PavedDrive01 indicates whether the driveway is paved, and GaragedCat reflects garage type. Finally, an interaction between OverallQual and living area was included to account for the idea that the value of construction quality depends on home size. Together, these variables provide a clear representation of the structural factors that influence property value.

Model Selection and Final Performance:

The final model focuses on structural features that drive home values, including living area, construction quality, age, lot size, total/finished basement space, garage area, driveway type, porch space, and garage classification. Furthermore, we included an interaction between quality and living area to reflect that higher-quality materials tend to add more value in larger homes. Overall, these predictors were chosen because they represent the main

characteristics that influence buyer demand and market pricing. During model trial and error, Adjusted R^2 was used to guide model selection. Since adjusted R^2 accounts for the number of predictors included, it helps ensure that added variables genuinely improve the model. The final model was selected because it produced the highest Adjusted R^2 while remaining accurate and interpretable. The final results show an R^2 of 0.929 and an Adjusted R^2 of 0.928, meaning about 93% of the variation in sale prices is explained by the model. The RMSE of 0.1011 represents an average prediction error of about 10%, indicating strong and reliable pricing accuracy.

Interpretation of Model Coefficients:

Since the dependent variable is the natural logarithm of SalePrice, most coefficients represent approximate percentage changes in price, assuming all other variables are constant. The intercept of 9.4243 represents the baseline predicted log price when all numeric predictors are zero. OverallQual has a coefficient of −0.2598, but because the model includes an interaction with log_GrLivArea, this value is not interpreted alone. The positive interaction means that as living area increases, the overall effect of quality becomes positive, so higher-quality construction increases price for larger sized homes. The coefficient on log_GrLivArea is 0.1168, meaning a 1% increase in living area raises price by about 0.117%, before accounting for the interaction. Age has a coefficient of −0.0029, so each additional year reduces price by roughly 0.29%. A 1% increase in lot size, labeled log_LotArea, increases price by about 0.11%. TotalBsmtSF and GarageArea coefficients imply about a 0.01% increase in price per additional square foot, while BsmtFinSF1 increases price by about 0.0085% per square foot. Homes with paved driveways sell for approximately 8.28% more, and those in the GaragedCat category sell for about 3.49% more than their reference groups. TotalPorchSF adds roughly 0.01% to price per additional square foot. Overall, size, lot area, garage features, and driveway quality increase value, while age decreases it, and quality generates stronger returns in larger homes.

Model Assessment:

The Adjusted R^2 of 0.928 indicates that the selected housing characteristics explain nearly 93% of the differences in sale prices across homes in the dataset. This suggests that the model captures the main drivers of value. The RMSE of approximately 10% indicates that prediction errors are relatively small for real estate applications. While some variation remains due to factors not included in the dataset, such as buyer preferences or neighborhood characteristics, the model provides a strong and dependable evaluation.

Example Prediction and Interpretation:

To demonstrate model use, consider a 34 year old home with 6 OverallQual, 1,456 square feet of living area, 9,452 square foot lot, 990 square foot basement space (384 finished), 480 square foot garage, 48 square foot porch, and a paved driveway. Using the model, the predicted log(SalePrice) is approximately 12.01, which converts to an estimated sale price of about $164,000. Given the model's average error of about 10%, a reasonable price range would be from $147,000 to $180,000. This portrays how the model can support pricing and development decisions in real estate.

Conclusion:

This analysis shows that interior quality and living area are the strongest influencers on property value in Ames. The interaction between size and quality shows that higher construction quality generates larger returns in larger homes. Additionally, age reduces value over time, while basement space, garage features, and paved driveways provide more value. Lot and porch size contribute positively but play a smaller role. All in all, with an Adjusted R^2 of 0.928 and strong predictive accuracy, the final model provides an effective and interpretable outline for real estate valuation and planning.

Appendix:

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                            OLS Regression Results
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Dep. Variable:           logSalePrice   R-squared:                       0.929
Model:                            OLS   Adj. R-squared:                  0.928
Method:                 Least Squares   F-statistic:                     494.8
Date:                Wed, 11 Feb 2026   Prob (F-statistic):               0.00
Time:                        00:41:14   Log-Likelihood:                 808.93
No. Observations:                 927   AIC:                            -1568.
Df Residuals:                     902   BIC:                            -1447.
Df Model:                          24
Covariance Type:            nonrobust
==================================================================================================
                         coef    std err          t      P>|t|      [0.025      0.975]
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const                  9.4243      0.382     24.652      0.000       8.674      10.175
OverallQual           -0.2598      0.061     -4.284      0.000      -0.379      -0.141
log_GrLivArea          0.1168      0.052      2.268      0.024       0.016       0.218
Age                   -0.0029      0.000    -11.321      0.000      -0.003      -0.002
log_LotArea            0.1114      0.011     10.394      0.000       0.090       0.132
TotalBsmtSF            0.0001   1.51e-05      6.742      0.000    7.22e-05       0.000
BsmtFinSF1          8.461e-05   9.34e-06      9.055      0.000    6.63e-05       0.000
GarageArea             0.0001   2.33e-05      6.020      0.000    9.45e-05       0.000
PavedDrive01           0.0828      0.015      5.508      0.000       0.053       0.112
TotalPorchSF           0.0001   3.74e-05      2.838      0.005    3.27e-05       0.000
GaragedCat             0.0349      0.009      3.997      0.000       0.018       0.052
...
strong multicollinearity problems or that the design matrix is singular.

Final Adjusted R-squared: 0.9275
Root Mean Squared Error (RMSE): 0.1011
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